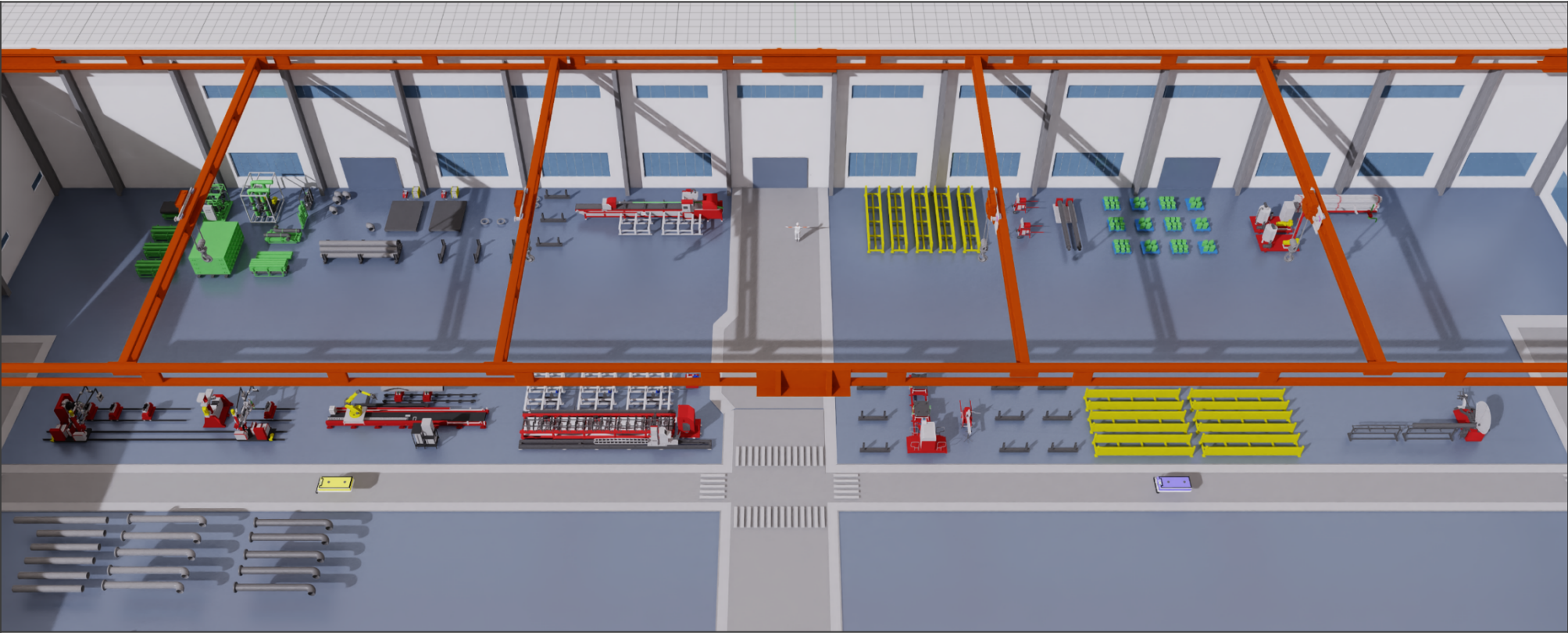
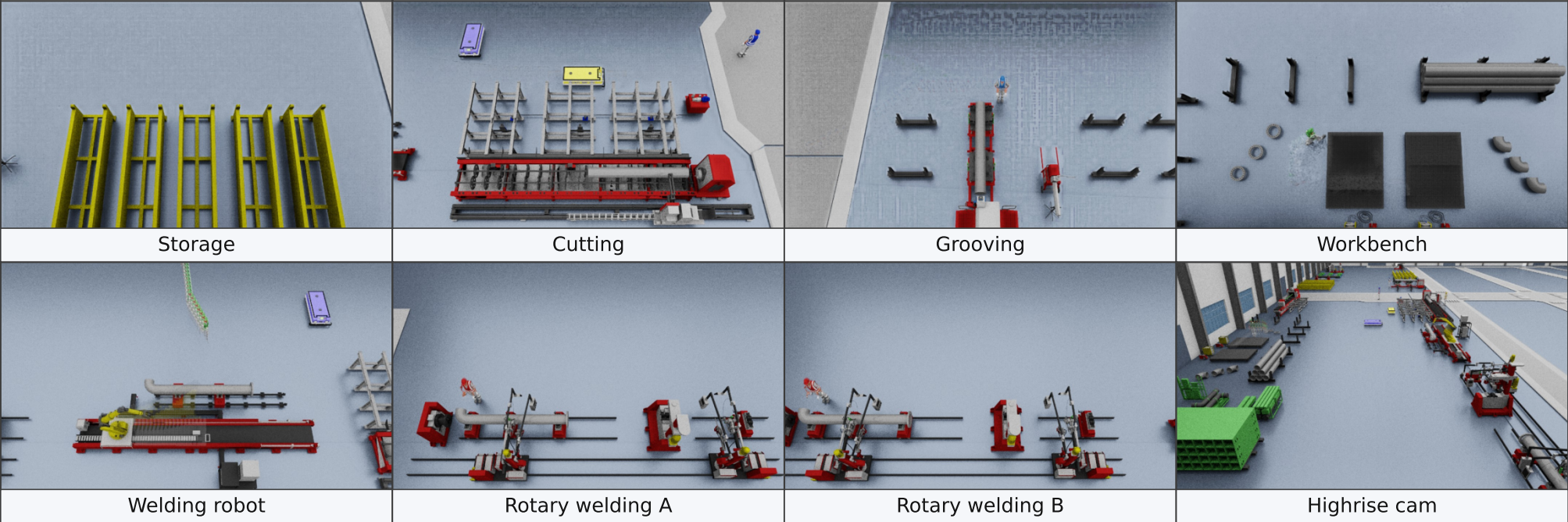


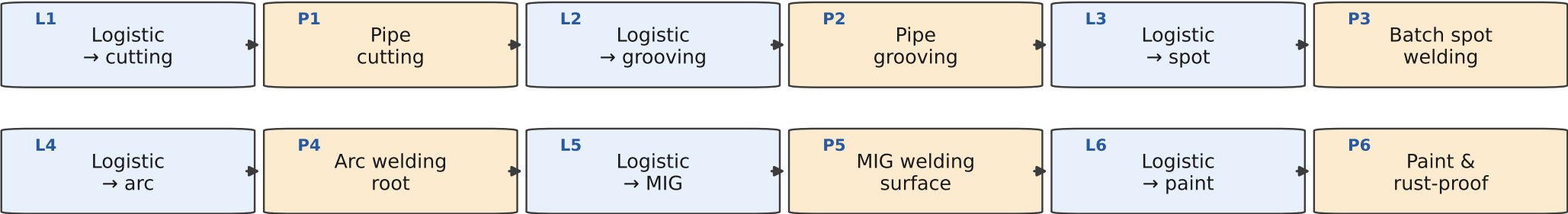
(a) Factory overview



(b) Multi-camera views



(c) Process tasks and exemplar subtask rows



L4

Logistic
→ arc

→

P4

Arc welding
root

→

L5

Logistic
→ MIG

→

P5

MIG welding
surface

→

L6

Logistic
→ paint

→

P6

Paint &
rust-proof

Logistic task

Processing task

Exemplar: logistic_for_pipe_cutting

Row	Human	Gantry	Machine	Robot
1	go_to_material	go_to_material	wait	go_to_material
2	material_on_gantry	wait	wait	wait
3	control_gantry	carry_to_robot	wait	wait
4	material_on_robot	wait	wait	wait
5	go_to_goal_area	move_to_goal	wait	carry_to_goal

Exemplar: pipe_cutting (processing)

Row	Human	Gantry	Machine
1	go_to_machine	none	wait
2	control_machine	none	process
3	wait	find_gantry	wait
4	control_gantry	go_to_machine	wait
5	material_on_gantry	wait	wait